

- 3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let $INCOME$ be household income (thousands of dollars) and $INSURANCE$ the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

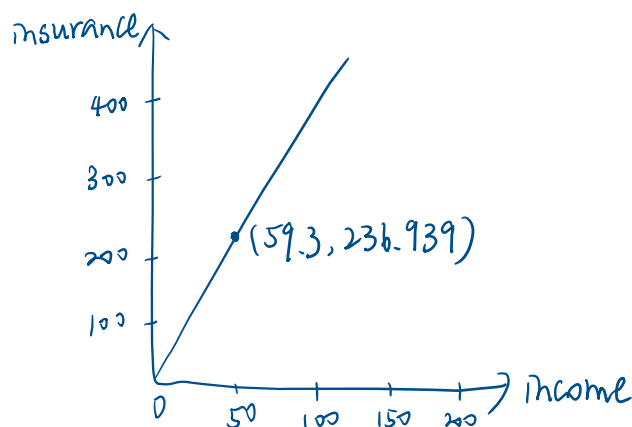
$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

- Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.
- Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

(a) regression pass through $(\bar{X}, \bar{Y})$

$$\widehat{insurance} = 6.855 + 3.88 \times 59.3 = 236.939$$



(b)

$$3.88 \times 1000 = 3880$$

⇒ Average amount of insurance held will increase 3880 with each additional 1000 of household income.

$$b_2 = 3.88, \alpha = \frac{0.05}{2} = 0.025, df = 18, t_c = 2.101$$

$$CI: [b_2 - t_c \times se(b_2), b_2 + t_c \times se(b_2)] \approx [3.645, 4.115]$$

⇒ In repeated sampling, about 95% intervals constructed (3.645, 4.115) will contain the true value of parameter β_2 .

$$(c) \text{ insurance} = 6.855 + 3.88 \times 100 = 394.855$$

$$\begin{aligned} \sqrt{\text{Var}(\hat{y})} &= \sqrt{\text{Var}(b_1) + 100^2 \text{Var}(b_2) + 2 \times 100 \times \text{Cov}(b_1, b_2)} \\ &= \sqrt{7.383^2 + 10000 \times 0.112^2 + 200 \times (-0.746)} \\ &= 5.545 \end{aligned}$$

$$\alpha = \frac{0.01}{2} = 0.005, df = 18, t_c = 2.878$$

$$CI: [394.855 - 2.878 \times 5.545, 394.855 + 2.878 \times 5.545] \approx [378.90, 410.81]$$

⇒ In repeated sampling, about 99% intervals constructed (378.90, 410.81) will contain the true value of parameter.

$$(d) \begin{cases} H_0: \beta_2 = 5 \\ H_1: \beta_2 \neq 5 \end{cases}, \text{ test statistic: } t = \frac{3.88 - 5}{0.112} = -10$$

$$\alpha = \frac{0.05}{2} = 0.025, df = 18, t_c = 2.101$$

$$\text{reject region: } \{t: t > 2.101\}$$

$$|t| = |-10| > t_c = 2.101 \rightarrow \text{reject } H_0$$

⇒ The data do not support the claim.

$$(e) \begin{cases} H_0: \beta_2 = 1 \\ H_1: \beta_2 > 1 \end{cases}, \text{ test statistic: } t = \frac{3.88 - 1}{0.112} = 25.71$$

$$\alpha = 0.01, df = 18, t_c = 2.552$$

reject region: $\{t: t > 2.552\}$

$$t = 25.71 > t_c = 2.552 \rightarrow \text{reject } H_0$$

$\Rightarrow$ It indicates that the slope is larger than 1.

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

- Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude? $\rightarrow SQFT = 20$
- Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. $\rightarrow SQFT = 40$

Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?

- Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.
- Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

(a)

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> summary(model)

Call:
lm(formula = price ~ I(sqft^2), data = collegetown)

Residuals:
    Min       1Q   Median       3Q      Max
-383.67  -48.39   -7.50   38.75  469.70

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  93.565854    6.072226   15.41  <2e-16 ***
I(sqft^2)     0.184519    0.005256   35.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.08 on 498 degrees of freedom
Multiple R-squared:  0.7122,    Adjusted R-squared:  0.7117
F-statistic: 1233 on 1 and 498 DF,  p-value: < 2.2e-16
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$$\hat{PRICE} = 93.5659 + 0.1845 SQFT^2$$

$$\begin{cases} H_0: \text{marginal effect} \leq 13 \\ H_1: \text{marginal effect} > 13 \end{cases}$$

$$\text{Marginal Effect} = \frac{d PRICE}{d SQFT} = 2 * SQFT * \hat{\alpha}_2 = 2(20)\hat{\alpha}_2 = 7.3807$$

$$se = \sqrt{\hat{var}(2SQFT\hat{\alpha}_2)} = 2 * SQFT * se(\alpha_2) = 0.2102$$

$$t\text{-value} = -26.7277, \quad p\text{-value} = 1$$

The right-tail critical value for the 5% level of significance is $t_{(0.95, N-2=492)} = 1.648$

rejection region $\{t : t > 1.648\}$, $t\text{-value} < t_c = 1.648$

This falls in the non-rejection region. We cannot conclude that the marginal effect is greater than \$13,000.

(b)

$$\hat{PRICE} = 93.56585 + 0.18452 SQFT^2$$

$$\begin{cases} H_0: \text{marginal effect} \leq 13 \\ H_1: \text{marginal effect} > 13 \end{cases}$$

$$\text{Marginal effect is } \frac{d PRICE}{d SQFT} = 2 * SQFT * \hat{\alpha}_2 = 2(40)\hat{\alpha}_2 = 14.7615$$

$$se = \sqrt{\hat{var}(2SQFT\hat{\alpha}_2)} = 2 * SQFT * se(\alpha_2) = 0.4205$$

$$t\text{-value} = 4.1893, \quad p\text{-value} = 1.65506723271392e-05$$

The right-tail critical value for the 5% level of significance is $t_{(0.95, N-2=492)} = 1.648$

rejection region $\{t : t > 1.648\}$, $t\text{-value} > t_c = 1.648$

This falls in the rejection region, which means there is an evidence that the marginal effect is greater than \$13,000.

(c)

$$\hat{E}(PRICE|SQFT = 20) = 93.5659 + 0.1845 * 20^2 = 167.3735$$

The standard error is

$$\sqrt{\text{var}(\hat{\alpha}_1) + 400 * (\hat{\alpha}_2)} = \sqrt{\text{var}(\hat{\alpha}_1) + 400^2 * (\hat{\alpha}_2) + 2(400)\text{cov}((\hat{\alpha}_1), (\hat{\alpha}_2))} = 4.746378$$

$$t_{(0.975, 498)} = 1.964739$$

The resulting interval estimate is $167.3735 \pm (1.964739) 4.746378 = [158.0481, 176.6988]$

In repeated sampling, we estimate with 95% confidence that the average price of a 2000 square foot house is between \$158,048.1 and \$176,698.8

(d)

In the sample there are 3 houses with 2000 square feet ($SQFT = 20$). They sold for \$138,000, \$169,000 and \$183,000 respectively, and the sample average is 163.3, which is inside the interval estimate $[158.0481, 176.6988]$.

3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable $SAL1$ = unit sales of brand no. 1 canned tuna, and $APR1$ = price per can of brand no. 1 tuna (in dollars).

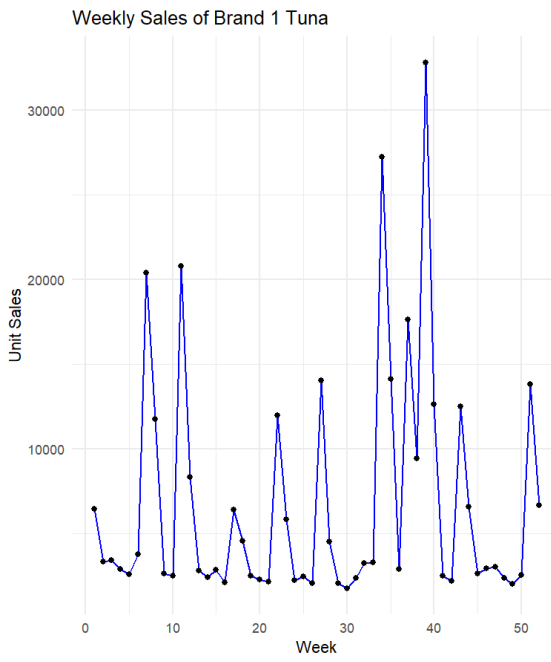
- a. Calculate the summary statistics for $SAL1$ and $APR1$. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus $WEEK$. How much variation in sales and price is there from week to week?
- b. Plot the variable $SAL1$ (y-axis) against $APR1$ (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?
- c. Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?
- d. Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.
- e. Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 “at the means.” Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?
- f. Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p -value for the test. What is your conclusion?

(a)

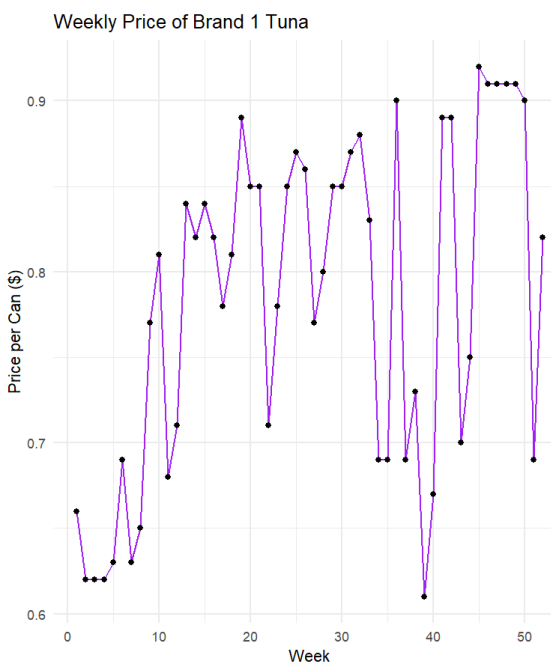
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> summary(tuna$sa11)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1762   2485   3136   6719   8612  32820

> summary(tuna$apr1)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.6100  0.6900  0.8100  0.7825  0.8625  0.9200

> sd(tuna$sa11)
[1] 6915.083
> sd(tuna$apr1)
[1] 0.09803711
```



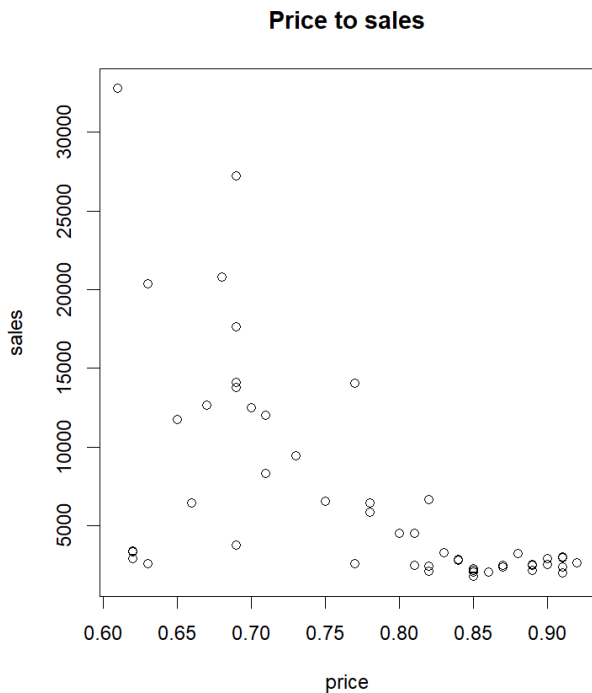
Weekly sales show strong fluctuations, with certain weeks experiencing sharp spikes. These peaks are most likely driven by promotional campaigns or seasonal demand.



The price per can remains relatively stable, ranging between \$0.60 and \$0.90. However, occasional drops in price align with promotional periods.

=> The graphs indicate that when the price decreases, sales volumes rise significantly, which means consumers are highly price-sensitive, and promotional discounts have a strong positive impact on demand.

(b)



We can see that when the price per canned tuna rises, the sales go down, which is a negative relationship.

It is expected because of law of demand.

When price is higher, the demand becomes lower and the sales decrease; in contrast, when price is lower, the demand is higher and the sales increase.

(c)

```
> summary(model)

Call:
lm(formula = sal1 ~ PRICE1, data = tuna)

Residuals:
    Min       1Q   Median       3Q      Max
-10869.5 -1588.3  -499.3   1626.2  18607.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  40714.22   6196.42   6.571 2.82e-08
PRICE1       -434.45    78.58  -5.528 1.17e-06

(Intercept) ***
PRICE1      ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,    Adjusted R-squared:  0.367
F-statistic: 30.56 on 1 and 50 DF, p-value: 1.172e-06
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$$\hat{SAL1} = 40714.22 - 434.45 PRICE1 + e$$

The effect of a one cent increase in the price means the margin effect of PRICE, that

is $\frac{\partial SAL1}{\partial PRICE1} = \beta_2$. The point estimator

for β_2 is $b_2 = -434.45$ ✎

$$b_2 = -434.45, SE(b_2)=78.58, t_{\frac{0.05}{2}}(52 - 2)=2.009$$

$$95\% \text{ C.I. is } b_2 \pm SE(b_2) \cdot t_{\frac{\alpha}{2}}(n - 2) = [-592.2897, -276.6049] \quad \text{✎}$$

(d) $\widehat{SAL1} = 40714.22 - 434.45 \times 70 = 10,302.9$

Standard error of $\hat{Y}_i$ is $\sqrt{MSE \left(\frac{1}{n} + \frac{(X_i - \bar{X})^2}{S_{XX}} \right)} = 1,001.231$

Therefore, we can construct a 90% interval estimate by using

$\hat{Y}_i \pm t_{\alpha/2}(n-2) \times SE(\hat{Y}_i)$ to get $[8,624.934, 11,980.87]$

In repeated sampling, about 90% interval constructed $[8624.934, 11980.87]$ will contain the true value of the parameter.

(e)

$\text{mean}(PRICE1) = 78.25$

$\text{mean}(SAL1) = 6718.712$

$\epsilon = \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = m \frac{x}{y} = \beta_2 \frac{PRICE1}{SAL1} = -5.0598$

$se(\epsilon) = se(\beta_2) \times \frac{PRICE1}{SAL1} = 0.9152$

t-statistic $= t_{\frac{\alpha}{2}}(n-2)$, the interval is $\epsilon \pm t_{\frac{\alpha}{2}}(n-2) \times se(\epsilon) = 2.0086$

95% C.I. = $[-6.8981, -3.2215]$

$\epsilon = -5.0598$, sales are highly sensitive to price changes, indicating a high degree of elasticity.

Yes, canned tuna has many substitutes in the market. Consumers have plenty of alternatives to choose from if the price of canned tuna increases. So the degree of elasticity is high.

(f)

$H_0: \text{elasticity} = -3, H_1: \text{elasticity} \neq -3$

Test statistic: $\varphi = \text{elasticity} - (-3) / se(\text{elasticity})$

$\alpha = 0.1$, rejection region $\{|\varphi| : |\varphi| > t_{0.05}(50) = 1.6759\}$

$\varphi^* = (-5.059825 + 3)/0.9152 = -2.25 \in \text{rejection region}$

$p\text{-value} = 0.0288 < 0.1$, we reject the null hypothesis

There is evidence that the elasticity possibly does not equals to -3.